— VOICE AGENT LATENCY

Where *1.74 s* goes.

The median reply time of my AI phone agent over 20 simulated calls. Every stage, measured.

SWIPE →

— WHAT I TIMED

Caller stops talking. *Bot starts speaking.*

1.74 s
median
55 turns, 20 calls

4.61 s
p95
the slow tail

Measured on the server. The simulated caller measured almost the same: 1.80 s median, 4.63 s p95.

THE THREE STAGES

Medians, *stage by stage*

Speech-to-text 0.47 s

Deepgram, time to final transcript

LLM first token 0.87 s

gpt-4.1-mini

Voice first audio 0.24 s

Deepgram Aura-2

Medians of the parts don't add up to the median whole. The rest is the wait for the caller to finish, turn detection and sentence buffering.

BIGGEST PIECE

The LLM's first token: *0.87 s*

The largest single stage is the model starting to answer. It is the first place I look when a reply feels slow: prompt size, model choice, and whether a tool call sits in the turn.

— THE SLOW TAIL

1 turn in 20 takes *4.6 s* or more.

A median hides the turns callers remember. The call console shows a waterfall for every turn, so a slow one points at the stage that stalled.

```
turn latency · burst pipe call
endpointing wait 0.20
transcription 0.24
LLM inference 1.23
sentence buffer 0.32
speech synthesis 0.37
```

HONEST LIMITS

What these numbers *leave out*

- Calls were simulated over Twilio's Media Streams protocol, not dialed from a phone.
- So the phone network is not in the number. Real calls will be somewhat slower.
- The callers were an LLM following personas, not real people.

WHY MEASURE THIS WAY

Your own voice stack, *not a rented one.*

Own the pipeline and every stage is yours to time and tune.

See every turn: adelatya.com/voice-agent-engineer/phone-agent

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